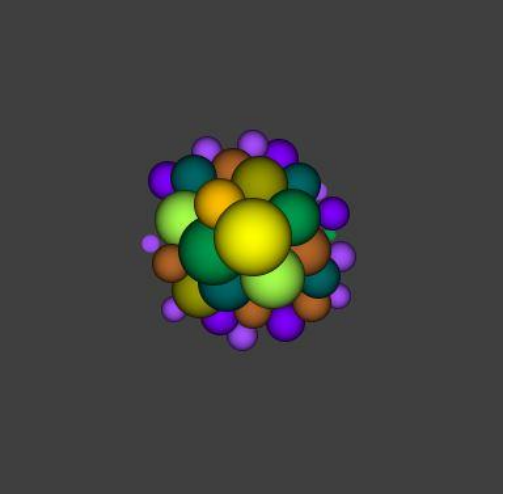
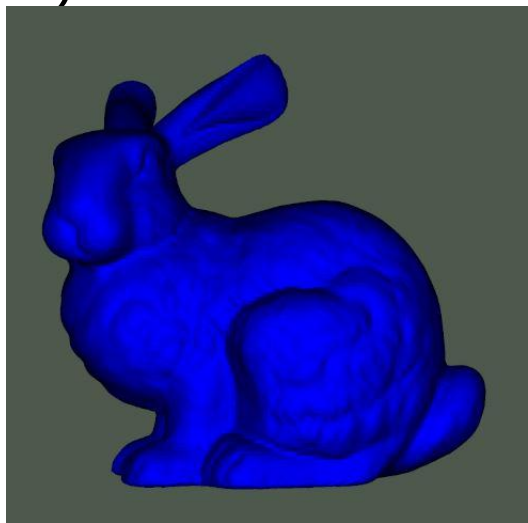


CS 440: Computer Graphics

Week 4

Lecture 7: Affine Geometry (Coordinate Systems and Frames)



Acknowledgements: Cem Yuksel, University of Utah; Waqar Saleem, Habib University

Announcements and Recap

- Quiz today
- Recitation this Friday?

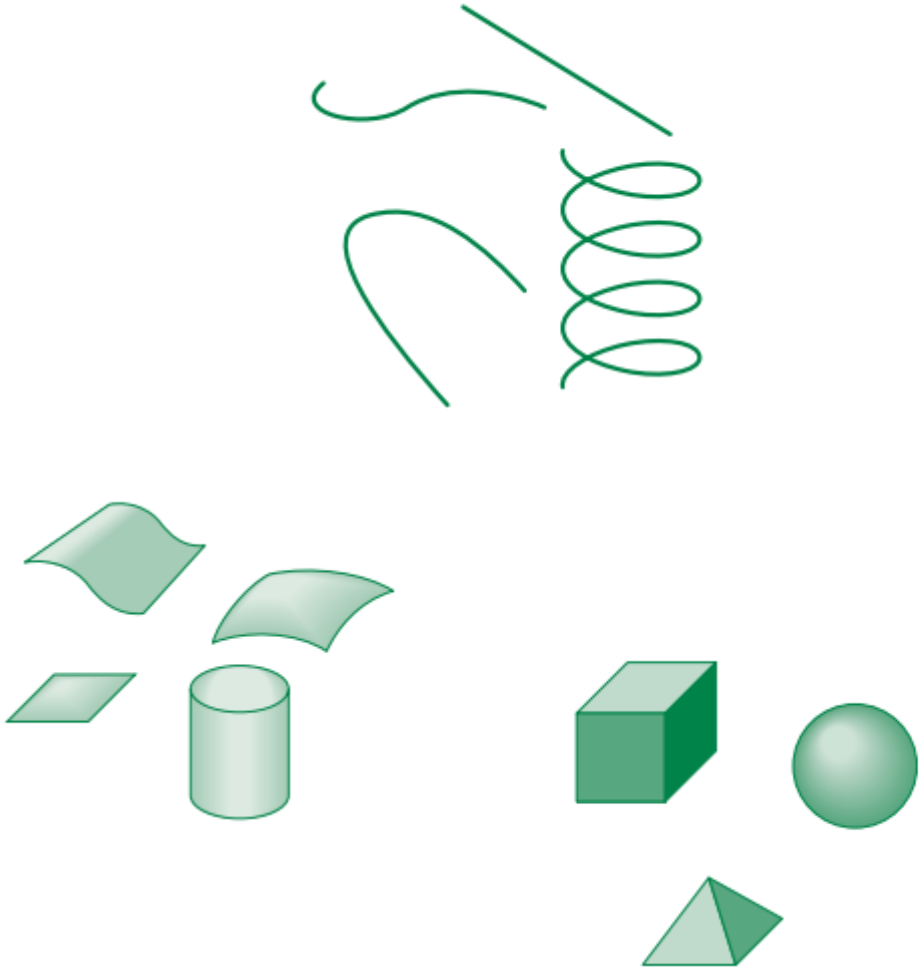
- Recap:
  - Affine Space
  - Affine Geometry: Lines and Planes
  - Barycentric Coordinates
  - Normals

Today: Affine Geometry

- 3D Primitives
- Coordinate System
- Change in Coordinate System
- Coordinate Frame
- Change in Coordinate Frame
- Homogenous Coordinates

3D Primitives

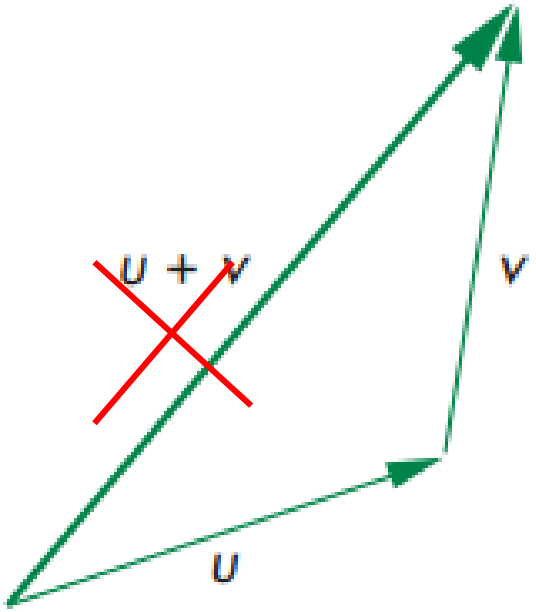
- Objects are hollow
- Defined by vertices
- Approximated by polygons



Short video on Tessellation:
https://youtu.be/7GiKeeWSf4s?si=_3bHi8v6B7dLlTlb

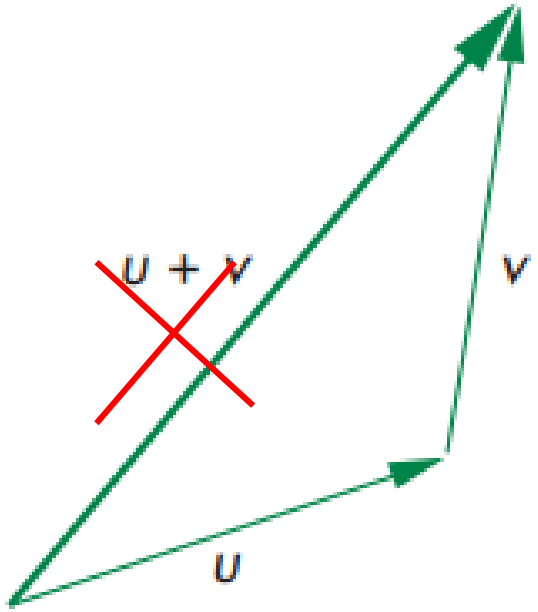
Linear Independence

- In a vector space, a set of vectors is **linearly independent** if we cannot write one of the vectors in terms of the others using the defined operations.



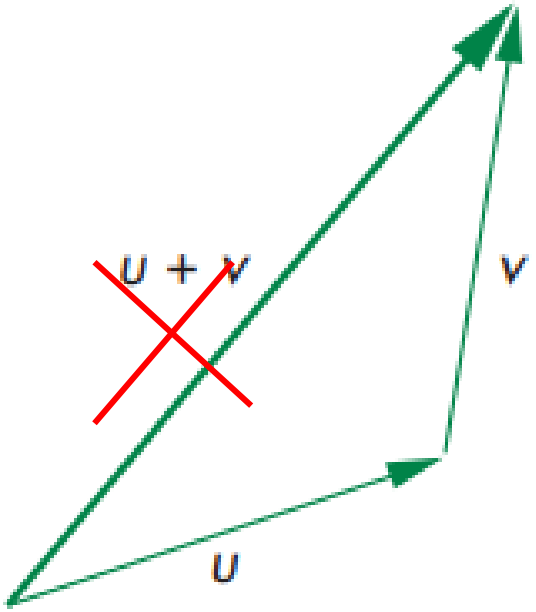
Dimension and Basis

- A vector space has a **dimension,** which is the maximum number of linearly independent vectors that we can find.



Basis

- In an n-D space, any set of n linearly independent vectors form a **basis** for the space.



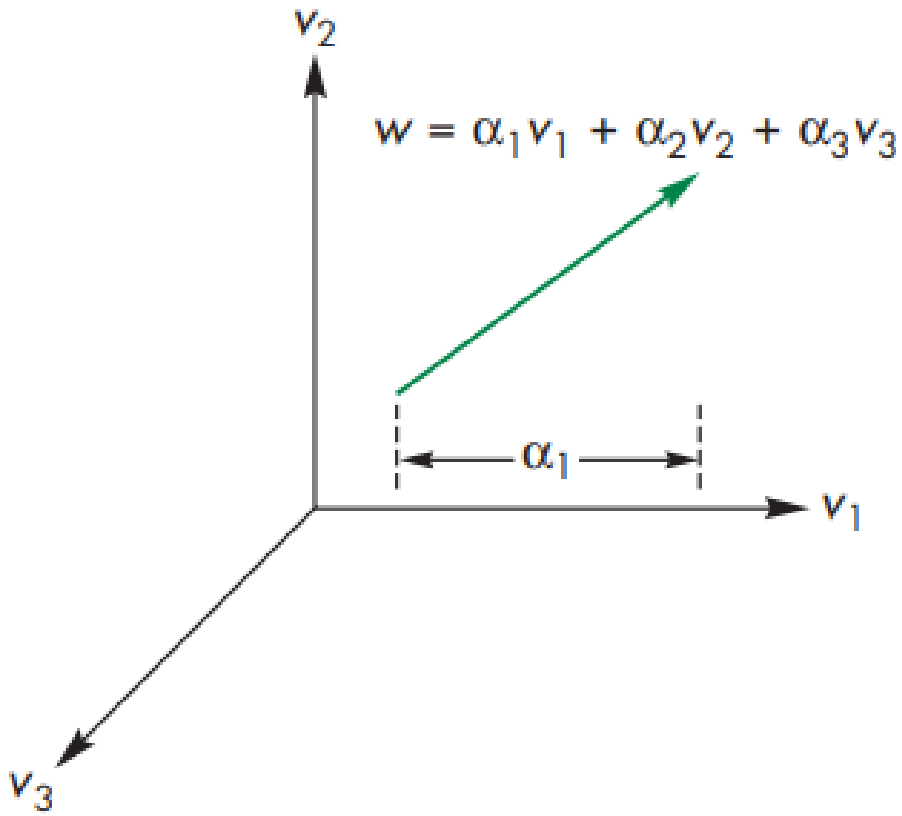
Coordinate System: Vector Representation

$$\vec{w} = \alpha_1 \vec{v_1} + \alpha_2 \vec{v_2} + \alpha_3 \vec{v_3}$$

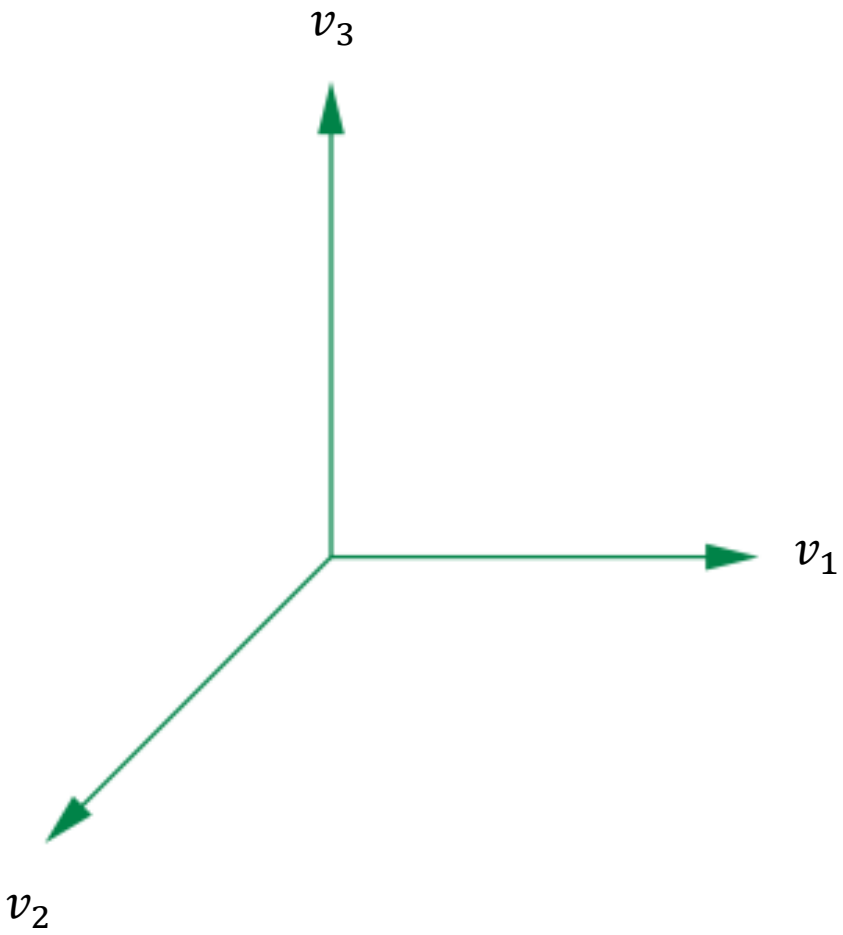
$$\vec{w} = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{bmatrix} \begin{bmatrix} \vec{v_1} \\ \vec{v_2} \\ \vec{v_3} \end{bmatrix}$$

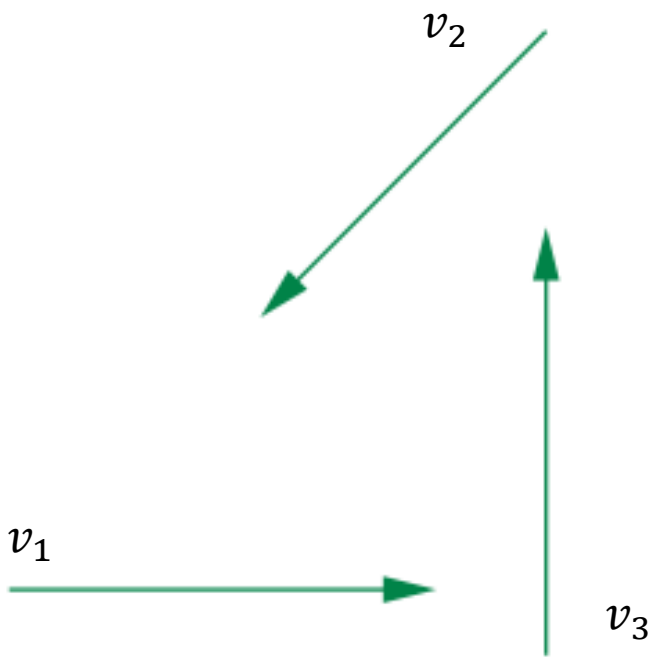
$$\vec{w} = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{bmatrix} \mathbf{v}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{v}$$



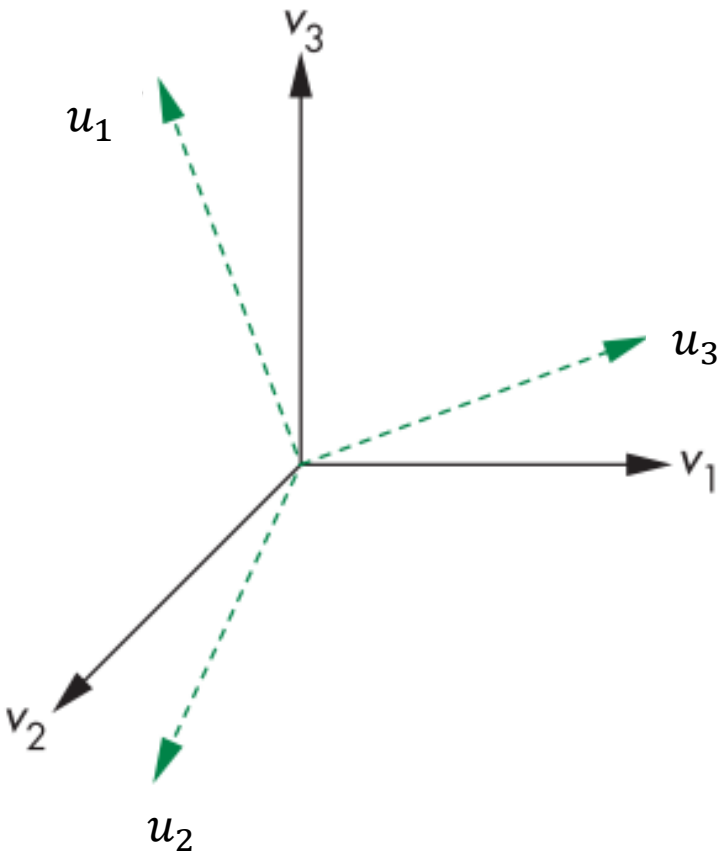
The Problem with Coordinate Systems





Change of Coordinate System: Basis

- What is the vector representation if we change the basis vectors?



Change of Coordinate System: Basis

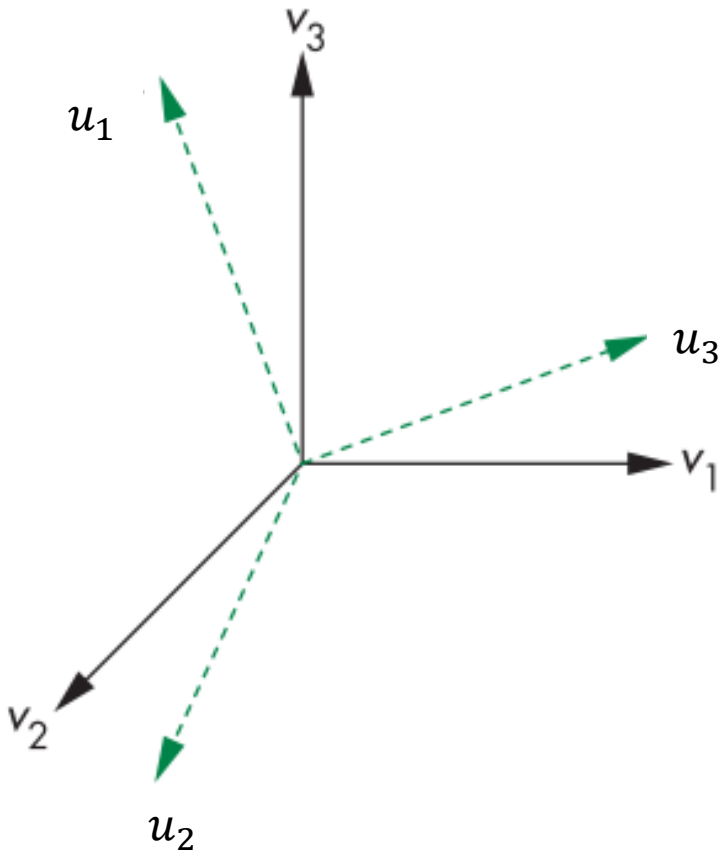
$$\overrightarrow{u_1} = \gamma_{11}\overrightarrow{v_1} + \gamma_{12}\overrightarrow{v_2} + \gamma_{13}\overrightarrow{v_3}$$

$$\overrightarrow{u_2} = \gamma_{21}\overrightarrow{v_1} + \gamma_{22}\overrightarrow{v_2} + \gamma_{23}\overrightarrow{v_3}$$

$$\overrightarrow{u_3} = \gamma_{31}\overrightarrow{v_1} + \gamma_{32}\overrightarrow{v_2} + \gamma_{33}\overrightarrow{v_3}$$

$$\mathbf{M} = \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} \\ \gamma_{31} & \gamma_{32} & \gamma_{33} \end{bmatrix}$$

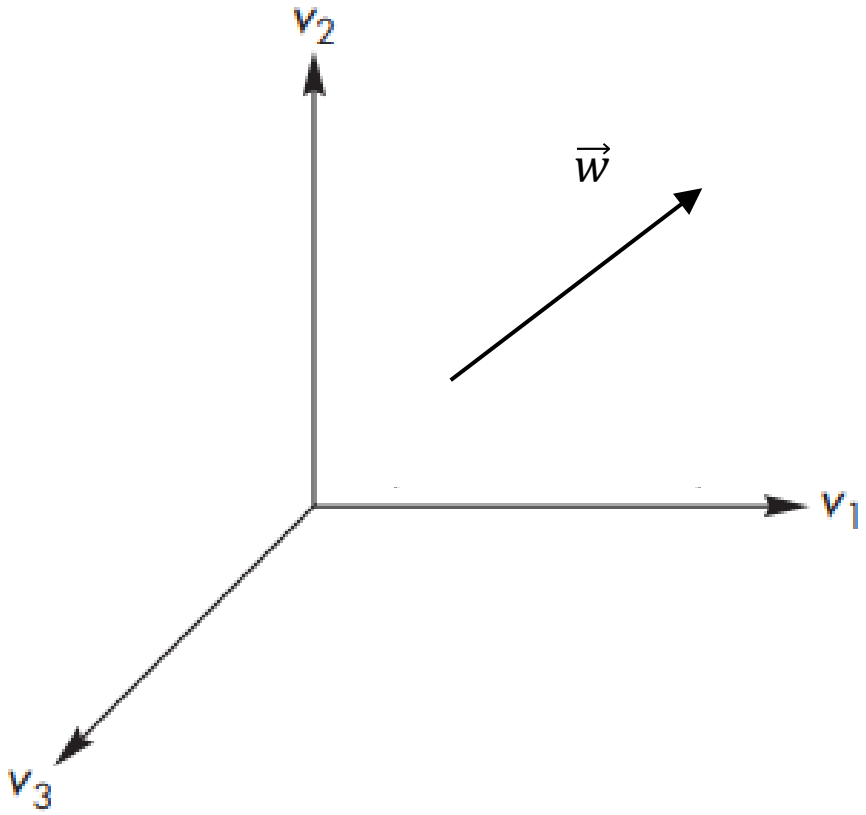
$$\mathbf{u} = \mathbf{M}\mathbf{v}$$



Coordinate System: Vector Representation

$$\vec{w} = \alpha_1 \overrightarrow{v_1} + \alpha_2 \overrightarrow{v_2} + \alpha_3 \overrightarrow{v_3}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{v}$$



Change of Coordinate System: Vector Representation

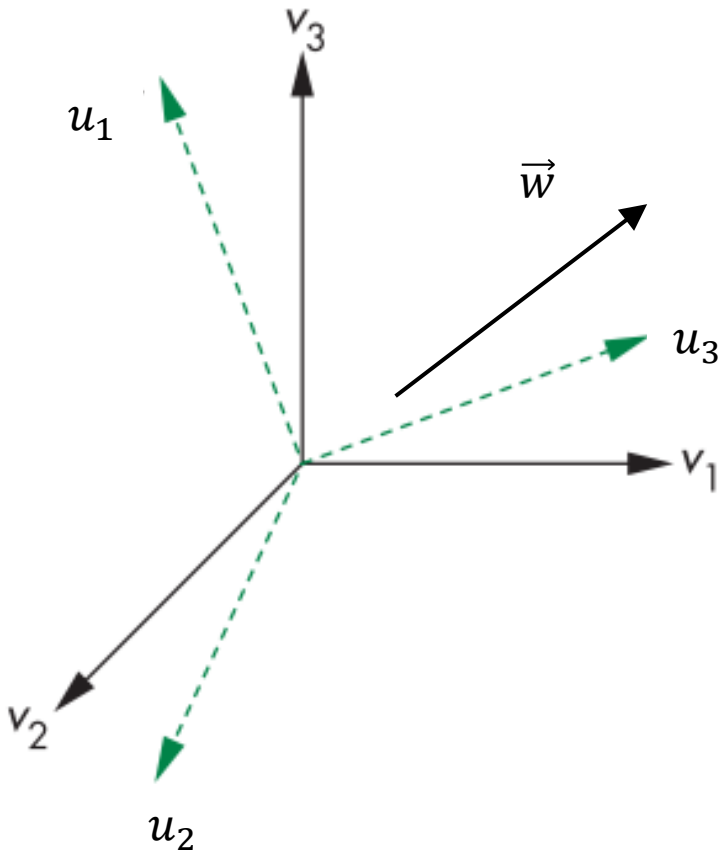
$$\vec{w} = \alpha_1 \overrightarrow{v_1} + \alpha_2 \overrightarrow{v_2} + \alpha_3 \overrightarrow{v_3}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{v}$$

$$\vec{w} = \beta_1 \overrightarrow{u_1} + \beta_2 \overrightarrow{u_2} + \beta_3 \overrightarrow{u_3}$$

$$\vec{w} = \mathbf{b}^\top \mathbf{u}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{v} = \mathbf{b}^\top \mathbf{u}$$



Change of Coordinate System: Vector Representation

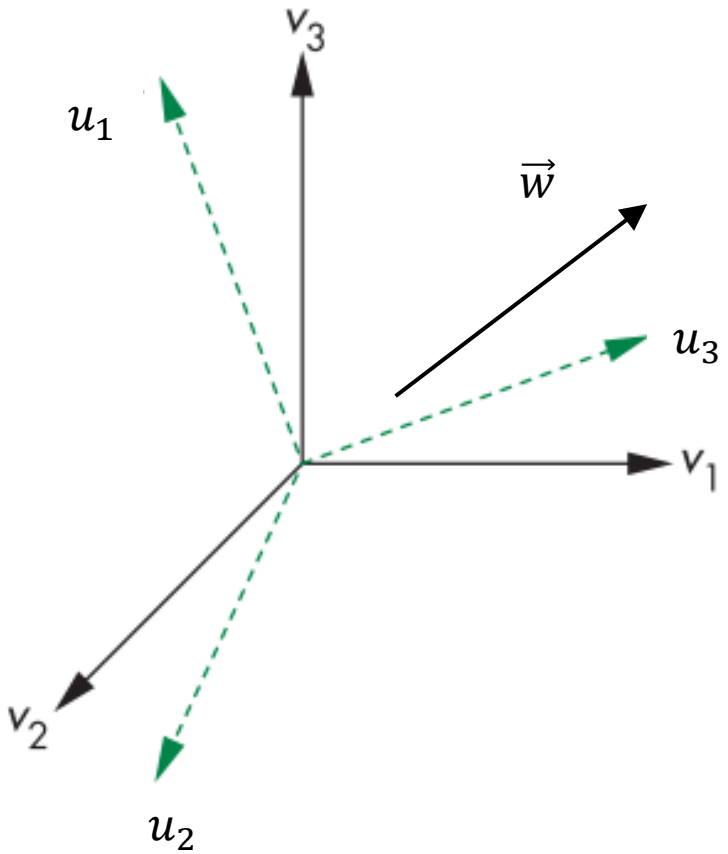
$$\vec{w} = \mathbf{a}^\top \mathbf{v} = \mathbf{b}^\top \mathbf{u}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{v} = \mathbf{b}^\top \mathbf{M}\mathbf{v}$$

$$\mathbf{b}^\top \mathbf{M} = \mathbf{a}^\top$$

$$\mathbf{M}^\top \mathbf{b} = \mathbf{a}$$

$$\mathbf{b} = (\mathbf{M}^\top)^{-1}\mathbf{a}$$

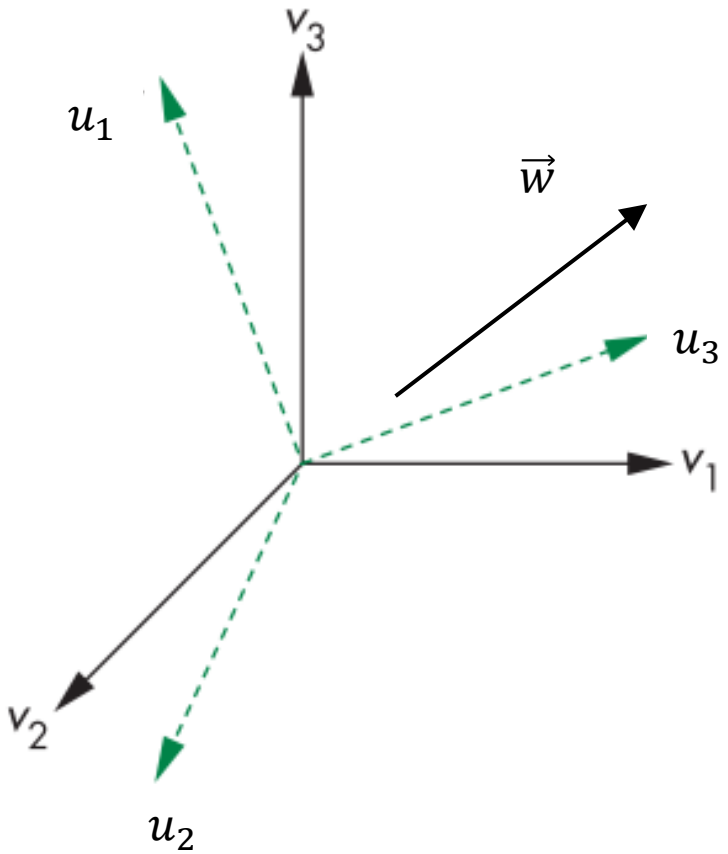


Change of Coordinate System: Vector Representation

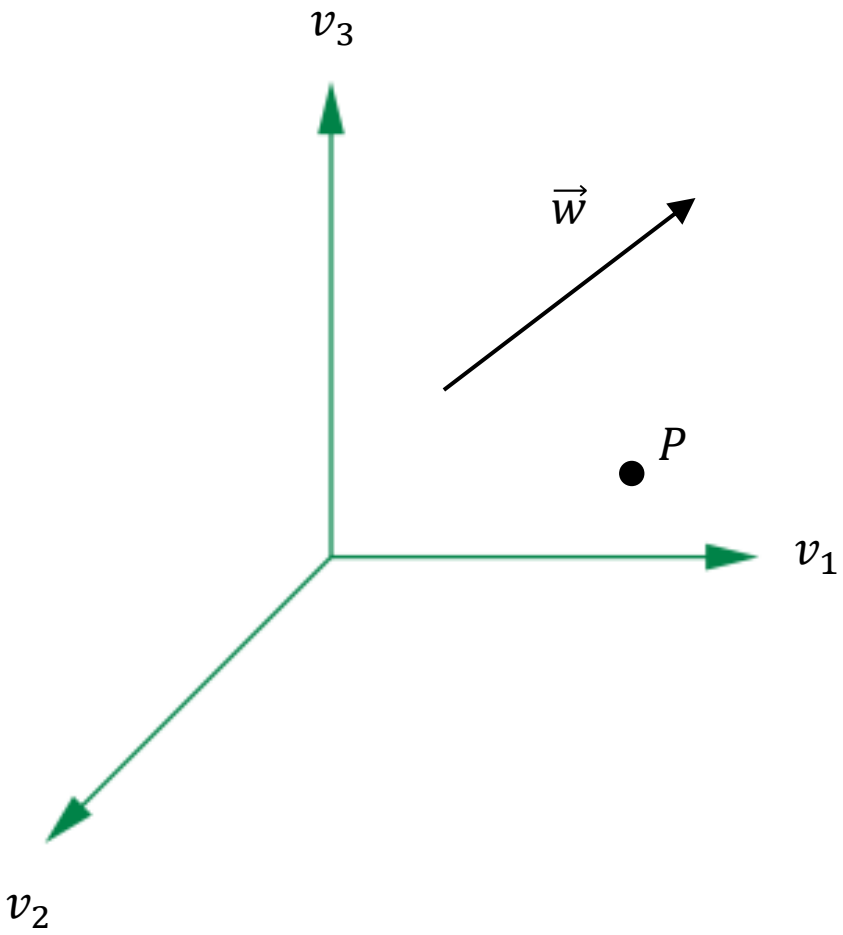
$$\vec{w} = \mathbf{a}^\top \mathbf{v}$$

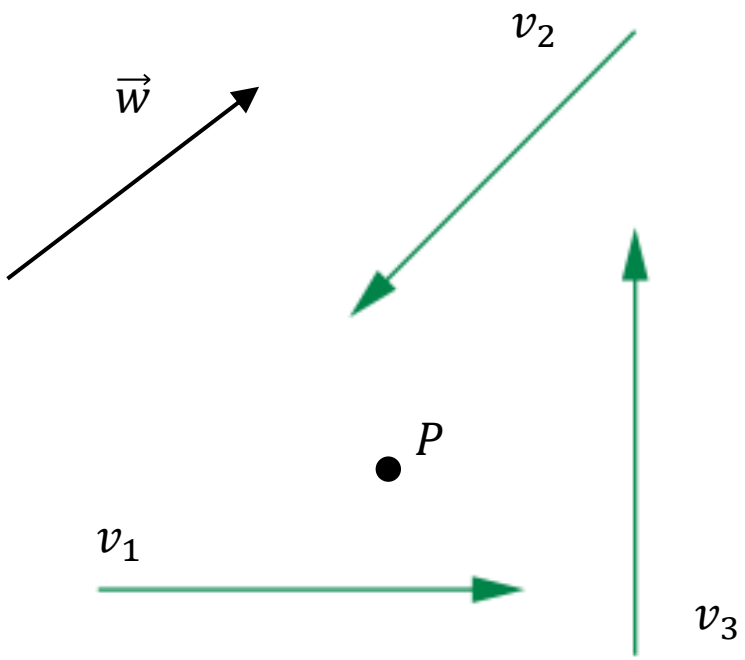
$$\mathbf{u} = \mathbf{M}\mathbf{v}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{M}^{-1}\mathbf{u}$$

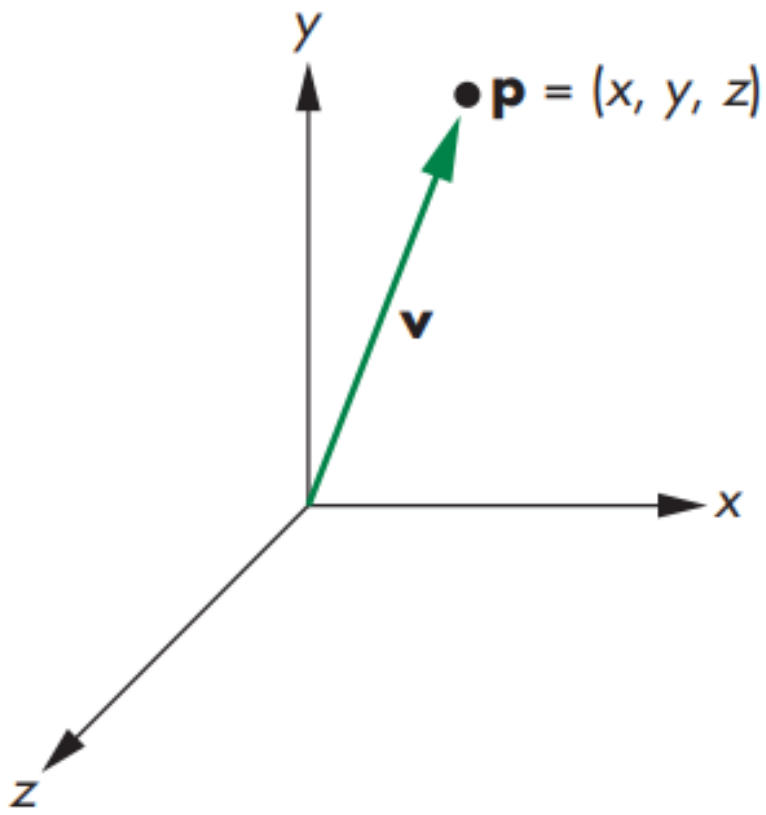


The Problem with Coordinate Systems





The Problem with Coordinate Systems



Coordinate Frame

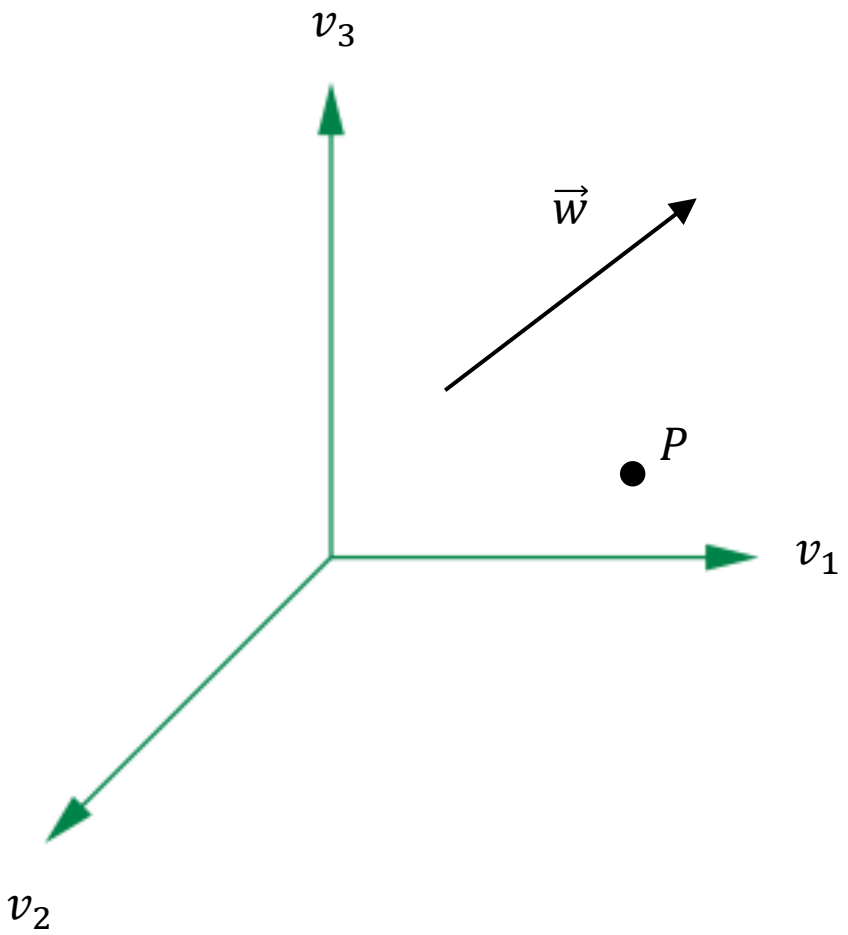
- Representation of points and vectors in an affine space

$$\vec{w} = \alpha_1 \overrightarrow{v_1} + \alpha_2 \overrightarrow{v_2} + \alpha_3 \overrightarrow{v_3}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{v}$$

$$P = P_0 + \beta_1 \overrightarrow{v_1} + \beta_2 \overrightarrow{v_2} + \beta_3 \overrightarrow{v_3}$$

$$P = P_0 + \mathbf{b}^\top \mathbf{v}$$



Coordinate Frame

- What if I "multiply" a point by 0 and 1?

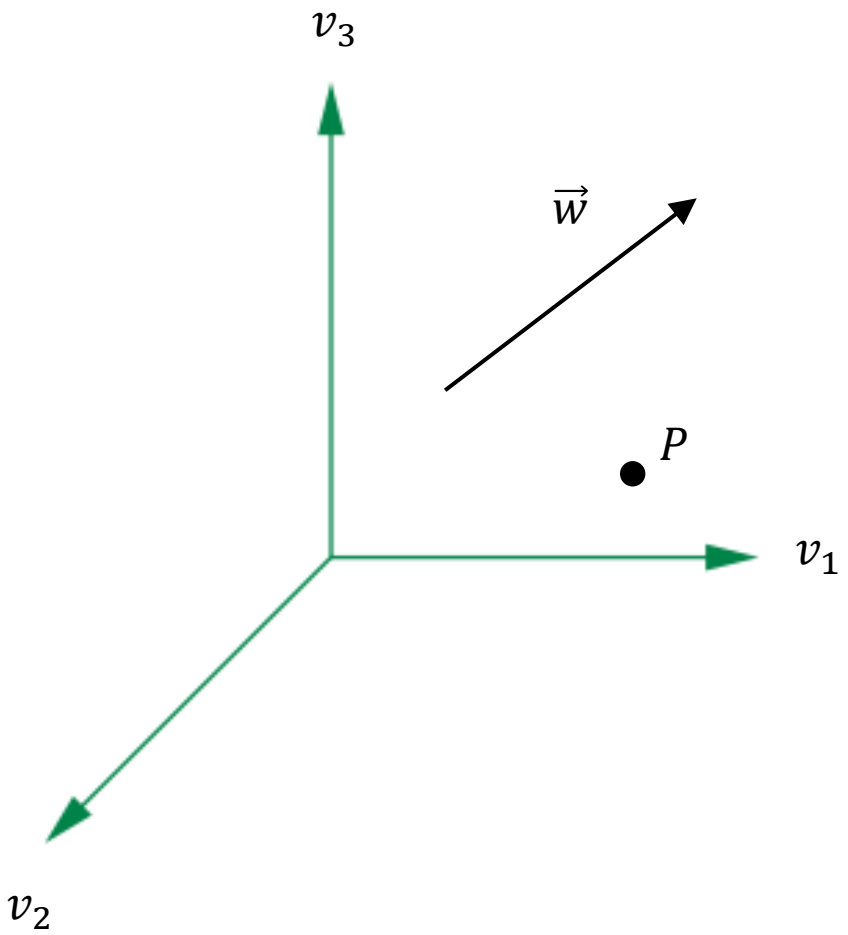
$$1.P = P$$

$$0.P = 0$$

- Point & Vector Representations:

$$\vec{w} = 0P_0 + \alpha_1\overrightarrow{v_1} + \alpha_2\overrightarrow{v_2} + \alpha_3\overrightarrow{v_3}$$

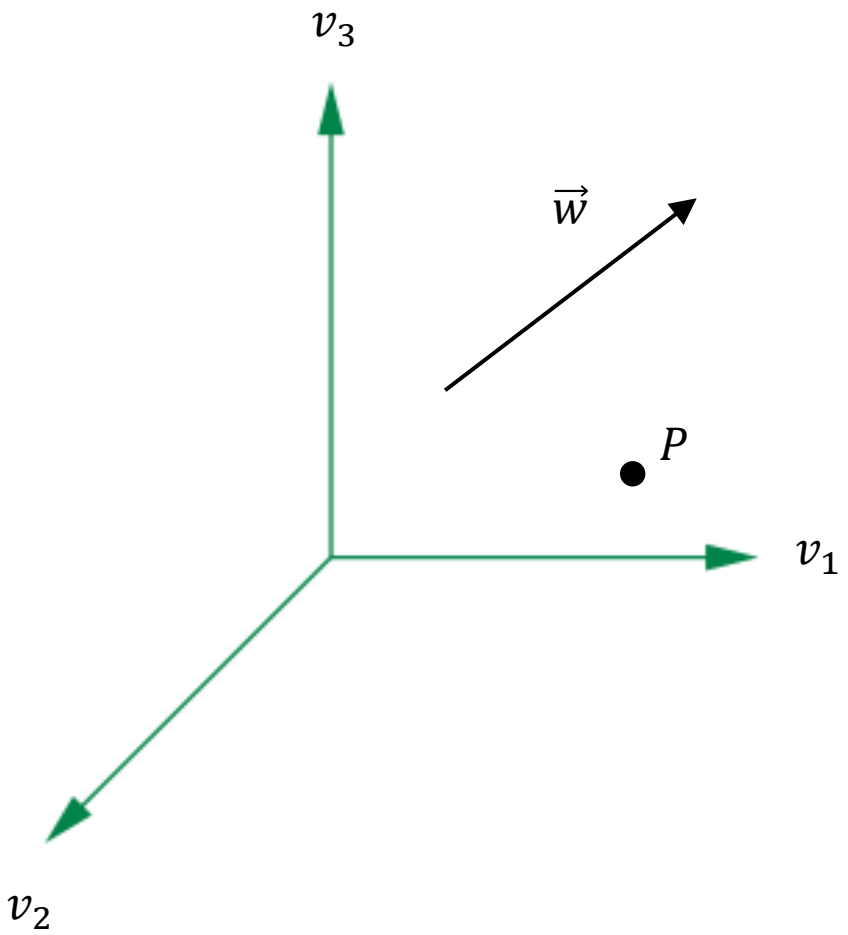
$$P = 1P_0 + \beta_1\overrightarrow{v_1} + \beta_2\overrightarrow{v_2} + \beta_3\overrightarrow{v_3}$$



Homogenous Coordinates

$$\vec{w} = [\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad 0]\begin{bmatrix} \overrightarrow{v_1} \\ \overrightarrow{v_2} \\ \overrightarrow{v_3} \\ P_0 \end{bmatrix}$$

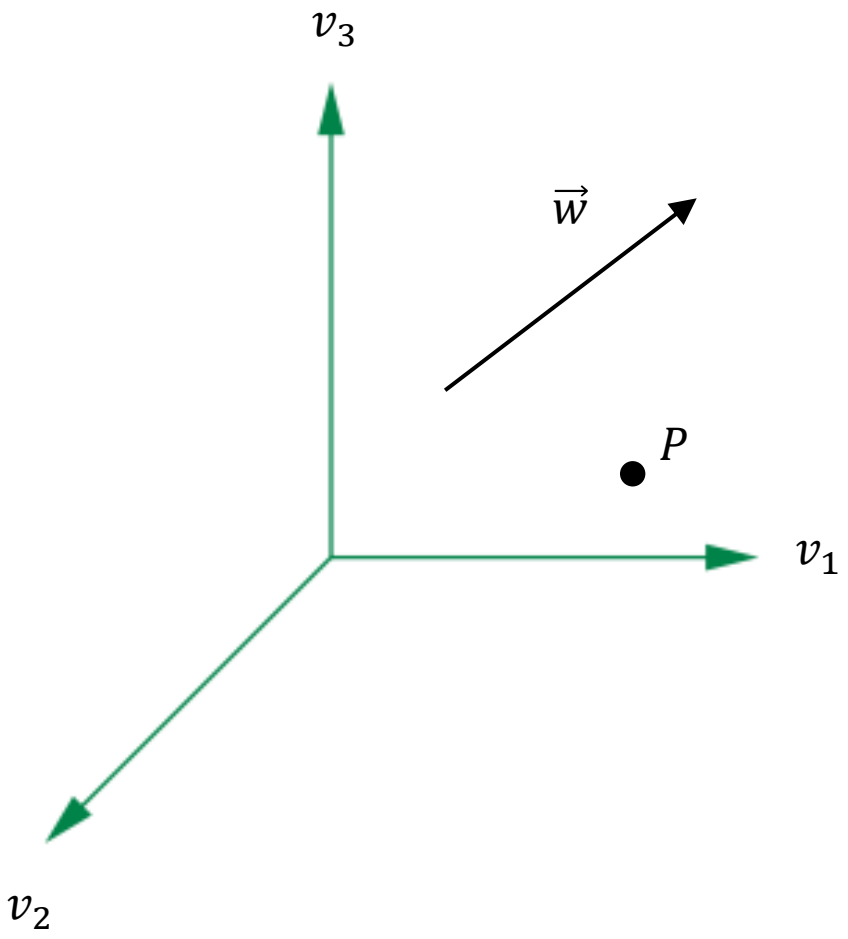
$$P = [\beta_1 \quad \beta_2 \quad \beta_3 \quad 1]\begin{bmatrix} \overrightarrow{v_1} \\ \overrightarrow{v_2} \\ \overrightarrow{v_3} \\ P_0 \end{bmatrix}$$



Homogenous Coordinates

$$\overrightarrow{w'} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ 0 \end{bmatrix}$$

$$P' = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ 1 \end{bmatrix}$$

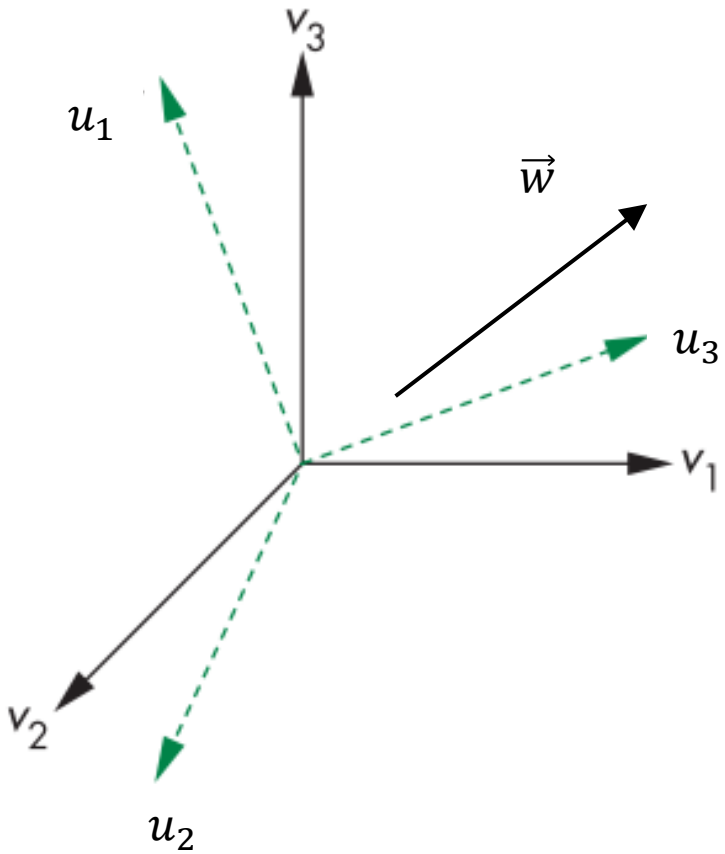


Recall: Change of Coordinate System

$$\vec{w} = \mathbf{a}^\top \mathbf{v} = \mathbf{b}^\top \mathbf{u}$$

$$\mathbf{u} = \mathbf{M}\mathbf{v}$$

$$\vec{w} = \mathbf{a}^\top \mathbf{M}^{-1}\mathbf{u}$$

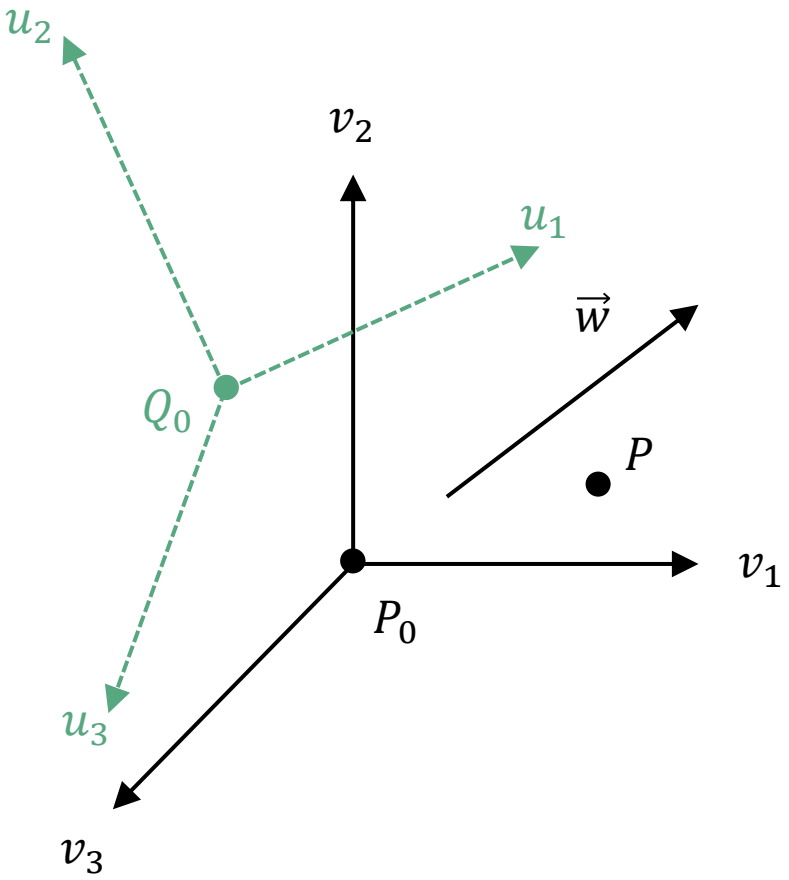


Change of Coordinate Frame

- Change of coordinate system from (v_1, v_2, v_3, P_0) and (u_1, u_2, u_3, Q_0)

$$\mathbf{u} = \mathbf{M}\mathbf{v}$$

$$\mathbf{M} = \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & 0 \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & 0 \\ \gamma_{31} & \gamma_{32} & \gamma_{33} & 0 \\ \gamma_{41} & \gamma_{42} & \gamma_{43} & 1 \end{bmatrix}$$



Change of Coordinate Frame

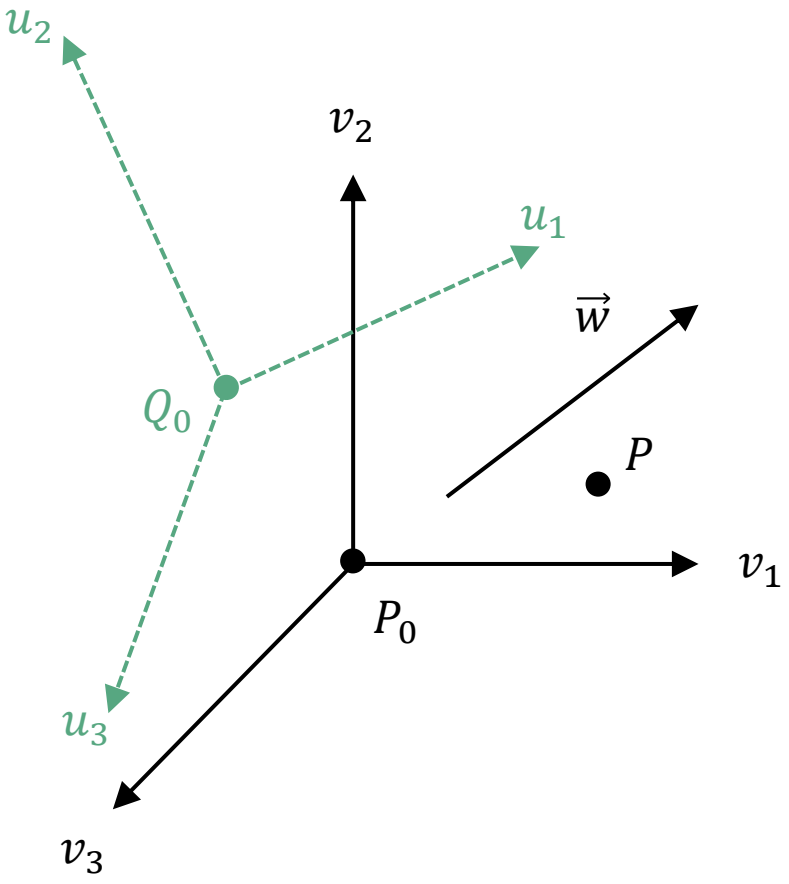
$$\vec{w} = \alpha_1 \vec{v_1} + \alpha_2 \vec{v_2} + \alpha_3 \vec{v_3} = \mathbf{a}^\top \mathbf{v}$$

$$P = \beta_1 \vec{v_1} + \beta_2 \vec{v_2} + \beta_3 \vec{v_3} = \mathbf{b}^\top \mathbf{v}$$

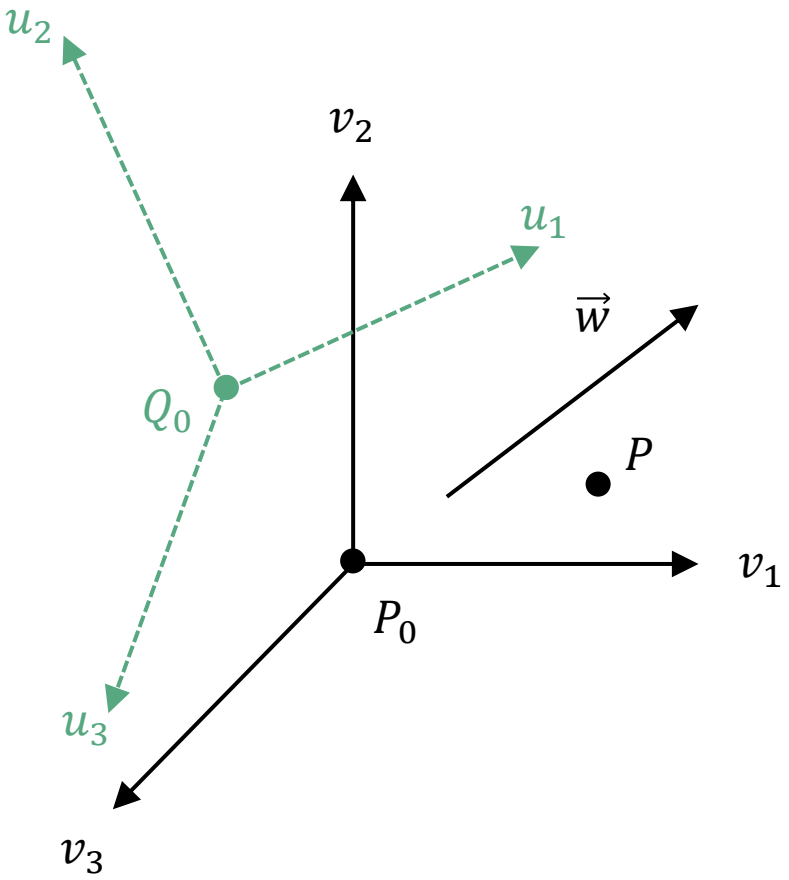
$$\mathbf{u} = \mathbf{M}\mathbf{v}$$

$$\vec{w} = (\mathbf{M}^\top)^{-1}\mathbf{a}\mathbf{u}$$

$$P = (\mathbf{M}^\top)^{-1}\mathbf{b}\mathbf{u}$$



Change of Coordinate Frame

- If we generalize homogenous coordinate representations of points and vectors…

$$\mathbf{a} = \mathbf{M}\mathbf{b}$$